2026/05/19

Quiz 5 Class 1 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

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1. (**T**) Only reentrant functions are safe to use within signal/interrupt handlers, other functions may cause data corruption. **(True/False, 2 pts)**

2. (**B**) A mutex can be used to prevent data races in a multi-threading environment. What is the reason? **(2 pts)**

A. A mutex ensures that codes within a mutex compiles to only one assembly instruction.
B. A mutex prevents multiple threads executing the same block of code at the same time.
C. A mutex marks a block of code as critical, so all its instructions run with root privilege.
D. A mutex changes codes within a mutex, stopping them from generating race conditions.

3. (**B**) In a computer with virtual machine installed, both the host kernel and the virtual machine kernel need to run with the highest privilege (Ring 0). What's the best option? **(2 pts)**

A. Force the virtual machine kernel to run at Ring 1, and host kernel runs in hypervisor mode.
B. The virtual machine kernel runs at Ring 0, and host kernel runs in hypervisor mode.
C. Force the host kernel to run at Ring 1, and virtual machine kernel runs in hypervisor mode.
D. The host kernel runs at Ring 0, and virtual machine kernel runs in hypervisor mode.

4. (Short Answer, 4 pts) Perform taint analysis for the source code below. Treat type casting as normal assignment.

```
print(untainted char msg);
tainted int *read_config();

int val = 48;

char identity(int *input) {
    return (char)(*input);
}

// Call 1
int *s1 = read_config();
char c1 = identity(s1);
print(c1);

// Call 2
int *s2 = &val;
char c2 = identity(s2);
print(c2);
```

Solution:
Use the name v for the global variable val. Use the names s1, s2, c1, and c2 for the variables in the main body. For the identity() function, use the names p and r for the parameter and return value respectively.

Global variable constraints:

v >= untainted

identity() function constraints:

r >= p

Call 1 constraints:

s1 >= tainted

p-1 >= s1

c1 >= r+1

untainted >= c1

Result: untainted >= c1 >= r+1 >= p-1 >= s1 >= tainted leads to illegal flow (untainted >= tainted).

Call 2 constraints:

s2 >= v

p-2 >= s2

c2 >= r+2

untainted >= c2

Result: untainted >= c2 >= r+2 >= p-2 >= s2 >= v >= untainted leads to legal flow (untainted >= untainted).